

# Shubham Acharya

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## Professional Summary

AI Engineer / Backend Developer with hands-on experience building production-oriented Generative AI applications and workflow automation systems using Python, FastAPI, LangChain, LangGraph, RAG, and LLM APIs. Experienced in designing multi-agent systems, dynamic AI workflow engines, modular plugin architectures, vector search pipelines, and scalable REST APIs. Proficient in PostgreSQL, Docker, Google Cloud, Supabase, OAuth/JWT authentication, and modern LLM ecosystems.

## Technical Skills

**Languages:** Python, SQL

**AI & Generative AI:** LangChain, LangGraph, RAG, AI Agents, Multi-Agent Systems, Prompt Engineering, Tool Calling, Vector Search, Embeddings

**Backend:** FastAPI, REST APIs, SQLAlchemy, Pydantic, OAuth 2.0, JWT

**Databases & Storage:** PostgreSQL, MySQL, MongoDB, ChromaDB, Supabase

**LLMs & AI Providers:** Google Gemini, Vertex AI, GPT, Claude, Groq, Llama

**Cloud & DevOps:** Docker, Docker Compose, Google Cloud Platform, Cloud Run, Vercel, Render

**Tools:** Git, GitHub, LangSmith

## Projects

### NexusAI — AI Workflow Automation & Research Platform

*Python • FastAPI • LangGraph • LangChain • Gemini • PostgreSQL • Docker*

Built a visual, node-based AI workflow platform enabling users to create and execute custom multi-step research and knowledge pipelines. Designed a modular plugin architecture using metadata, Pydantic schemas, and independent execution modules, allowing new AI nodes to be added without modifying the core execution engine. Developed a dynamic LangGraph execution engine that converts frontend workflow graphs into executable StateGraph workflows and resolves data dependencies between connected AI nodes. Implemented server-driven node discovery that dynamically scans backend plugins and exposes metadata and JSON Schemas for automatic frontend tool and configuration-panel generation.

### PrepMate AI — AI-Powered Interview Preparation Platform

*LangGraph • LangChain • FastAPI • Vertex AI • PostgreSQL • Docker*

Built an AI-powered interview preparation platform featuring resume analysis, ATS scoring, AI mock interviews, job search, and personalized career roadmaps. Designed LangGraph-based workflows for resume evaluation, interview question generation, candidate assessment, and career planning. Developed and deployed secure FastAPI services using Google OAuth, JWT authentication, PostgreSQL, Supabase, Docker, and Google Cloud.

### Medical AI Assistant — Retrieval-Augmented AI System

*LangGraph • LangChain • FastAPI • ChromaDB*

Developed a RAG-based medical information assistant with intelligent tool routing for evidence-backed responses. Integrated PubMed, OpenFDA, Tavily Search, and Wikipedia for multi-source information retrieval. Built a PDF ingestion pipeline using semantic chunking, embeddings, ChromaDB vector storage, and citation-aware response generation.

### AuraChat — Full-Stack RAG Chatbot

*FastAPI • React • PostgreSQL • Docker • ChromaDB*

Built a full-stack RAG application enabling users to upload documents and interact with them through conversational AI. Implemented semantic search, persistent chat history, JWT authentication, document management, and Docker-based deployment.

## Experience

### Digital Computer Instructor

*Dec 2025 – Mar 2026*

*BR Birla Group of Schools, Jodhpur*

Taught computer fundamentals and Python programming through practical, project-based classroom sessions. Designed hands-on exercises to explain programming concepts, problem-solving, and computational thinking.

### Computer Instructor

*Jul 2025 – Nov 2025*

*Parashar International School, Jodhpur*

Delivered Python programming and computer fundamentals to secondary school students. Simplified technical concepts through practical demonstrations and interactive learning activities.

## Education

### Bachelor of Computer Applications (BCA)

*2022 – 2025*

*Aishwarya College, Jodhpur*

**CGPA:** 7.0